# Evaluation of validation strategies and transferability of EEG attention decoding: a comparative analysis on open datasets

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## Abstract

EEG decoding models must generalize across people and experimental paradigms, yet non-stationary signals and the uncertain equivalence of labels such as "attention" make this difficult. Evidence on whether nominally aligned attention-related datasets provide useful zero-shot transfer across tasks and datasets remains limited. Here, we evaluated zero-shot generalization of a protocol-defined binary contrast between high-demand and low-demand conditions in two technically compatible public EEG datasets. The datasets were harmonized through shared 12-channel input, a fixed preprocessing pipeline, and a common label mapping. Drowsiness was excluded to avoid conflating the contrast with differences in arousal, visual input, and ocular activity. Five classifiers spanning linear, tree-based, and deep-learning approaches were assessed using group-disjoint within-dataset, cross-subject, cross-task, and cross-dataset validation, without target-specific adaptation. Mean balanced accuracy across decoders was nearly unchanged when generalizing to unseen participants within a dataset, decreasing from 65.5% to 65.4% in the mental attention-state dataset and from 69.1% to 67.1% in

SAM-40. In contrast, cross-task transfer was on average 6.4 percentage points below the corresponding target-specific baselines, and cross-dataset transfer was 13.7 percentage points below them. Performance declined in every eligible zero-shot transfer comparison. Absolute accuracy also varied materially across decoders. Thus, shared channels, technically compatible acquisition, and nominally aligned labels were insufficient to preserve target-specific performance under heterogeneous transfer. These leakage-resistant, directed pre-adaptation benchmarks support empirical source selection and provide matched references for evaluating whether transfer learning, domain adaptation, or personalization yields meaningful gains. The findings concern protocol-defined contrasts rather than a unitary neural marker of attention.

# Introduction

Electroencephalography (EEG) is a technique for recording brain activity using multichannel scalp electrodes that measure voltage signals reflecting underlying cognitive processes. It is widely used in neuroscience and brain–computer interface (BCI) research due to its non-invasive nature, high temporal resolution, portability and relatively low cost [1]. At the same time, EEG signals are inherently noisy, non-stationary, and exhibit high inter-subject and inter-session variability [2, 3]. In addition, these challenges are compounded by the scarcity of large, diverse, and well-annotated open EEG datasets [4]. Consequently, developing a robust decoding model with strong generalization capabilities across subjects, sessions and tasks remains an open and central challenge [2, 5]. One promising approach to solve these problems is transfer learning (TL). TL reuses representations already learned on one or more source problems to reduce the amount of labeled data and training effort a related target problem would otherwise require [6]. Its success depends not only on modeling and learning strategies but also on the degree of domain and label compatibility between source and target datasets.

For cognitive constructs such as attention, this compatibility is especially consequential and difficult to guarantee because attention has always been an ambiguous concept and is not a unitary construct with a single universally accepted operational definition [4, 7]. Rather, attention is an umbrella term encompassing a set of partially independent processes and functions, such as selecting relevant information, maintaining focus over time, shifting attention between tasks, and allocating cognitive resources. These processes may operate on both external sensory input and internally represented information, including information maintained in working memory [7–9]. The relative contributions of these processes may vary with task demands, sensory modalities, task goals, experimental instructions, assigned conditions, self-reported assessments, and other aspects of experimental design and label definition. Consequently, experimental paradigms described as measuring "attention" may probe non-equivalent configurations of cognitive processes. As a result, different datasets may represent different attention-related cognitive states while assigning them the same nominal labels.

Previous EEG transfer-learning research has been predominantly concerned with cross-subject and cross-session generalization, whereas cross-task, cross-dataset, and cross-paradigm transfer have received comparatively less systematic attention [2, 10, 11]. Other recent work has evaluated cross-subject robustness of deep decoders on multiple motor-imagery benchmark datasets while keeping the task paradigm fixed [12]. Consequently, systematic empirical evidence for source–target suitability across heterogeneous attention-related EEG datasets and tasks remains limited. Source–target suitability is often motivated conceptually or inferred from nominally aligned labels. This leaves unclear which source–target combinations provide useful zero-shot transfer, how much target discriminability is retained relative to target-specific baselines, and whether transferability is direction-dependent. Establishing such pre-adaptation baselines is necessary to justify source selection and to interpret gains or negative transfer from subsequent transfer-learning and domain-adaptation methods [13].

Here, we contribute to addressing this gap by systematically evaluating zero-shot generalization across two technically compatible public EEG datasets spanning multiple tasks and paradigms. We examine within-dataset baseline performance, cross-subject generalization, and zero-shot transfer across tasks and datasets under a common preprocessing pipeline, compare directed source–target pairs against target-specific baselines, and assess transferability using five decoders ranging from linear and tree-based baselines to deep neural networks. Together, these analyses yield reproducible pre-adaptation benchmarks for evaluating candidate sources and contextualizing both gains and negative transfer produced by subsequent transfer-learning and domain-adaptation methods.

# Open-access datasets

## EEG data for mental attention state detection (Dataset A)

The dataset described in [14] comprises 34 EEG recordings collected from five participants over the course of seven days, with the fifth participant completing only six sessions. During each session, participants operated a virtual train using the *Microsoft Train Simulator*. Each recording lasted approximately 35 to 55 minutes. The participant cohort consisted of healthy university student volunteers.

Each session followed the same predefined route in the virtual train simulator, which required little active control and kept task demands comparable across recordings. Each session was divided into three sequential phases, each designed to reflect a different target mental state: *focused*, *unfocused*, and *drowsy*. Participants followed standardized instructions from the experiment supervisor for each phase:

- **Focused (0–10 min):** The focused phase required active monitoring of and mental engagement with the simulator, including sustained attention to the virtual controls and screen events.
- **Unfocused (10–20 min):** During the unfocused phase, participants disengaged from the task, ceased interaction, and were instructed to keep their eyes open and remain awake.
- **Drowsy (20–30 min):** During the drowsy phase, participants were allowed to relax, close their eyes, and enter a drowsy or lightly dozing state, simulating reduced cognitive arousal.

EEG signals were recorded using a modified Emotiv headset, equipped with wet electrodes and supporting wireless Bluetooth transmission. The device recorded data at a sampling rate of 128 Hz and with a voltage resolution of 0.51 $\mu$V. The dataset includes 14 EEG channels, labeled according to the 10–20 international system.

## SAM-40: dataset of 40-subject EEG recordings (Dataset B)

The **SAM-40** dataset, described in [15], contains EEG recordings from 40 healthy participants (14 females and 26 males) with a mean age of 21.5 years. It was originally collected to investigate short-term stress responses induced by various cognitive tasks, which correspond to a *continuous cognitive engagement* paradigm as described in [16]. The experimental protocol included four conditions:

- **Resting state**: EEG recording during relaxation while listening to music.
- **Stroop Color-Word Test**: Participants were required to identify the color of the ink in which a word was printed, while ignoring the word's semantic meaning. The task included two types of stimuli: *congruent* (the color name matched the ink color) and *incongruent* (the color name and ink color differed). Each execution of the task included 11 stimuli.
- **Arithmetic problem solving**: Participants mentally solved a mathematical problem and determined whether the presented answer was correct, responding with a "thumbs up" or "thumbs down" gesture. Each execution of the task included 6 arithmetic stimuli.
- **Mirror Image Recognition**: Participants were shown images and asked to determine whether they were mirror-symmetric or asymmetric, responding with the same gestures. Each execution of the task included 8 images.

Each task lasted 25 seconds, with 5-second rest intervals and a 10-second instruction period preceding the next task. After the completion of each trial, participants rated the level of stress experienced for each task on a 10-point scale, where 1 indicated minimal stress and 10 indicated maximal stress. For each participant, three trials were recorded using different sets of stimuli.

EEG signals were recorded using an Emotiv Epoc Flex headset with 32 wet electrodes, labeled according to the international 10–20 system. Data were acquired at a sampling rate of 128 Hz and a voltage resolution of 0.51 $\mu$V. The 32 channels included in the dataset are: Cz, Fz, Fp1, F7, F3, FC1, C3, FC5, FT9, T7, CP5, CP1, P3, P7, PO9, O1, Pz, Oz, O2, PO10, P8, P4, CP2, CP6, T8, FT10, FC6, C4, FC2, F4, F8, and Fp2.

# Methodology

The methodological design of this study was developed to evaluate the generalizability and transferability of EEG-based attention decoding across different recording conditions, participants, and task paradigms. To this end, we adopted two publicly available datasets that differ in experimental protocols but share compatible technical specifications, thereby enabling a controlled comparative analysis. The datasets were harmonized through a binary operational labeling scheme that distinguishes conditions of high protocol-defined task demand from conditions of rest or task disengagement, enabling consistent evaluation across scenarios. This section outlines the rationale for dataset alignment, the strategies employed for cross-validation, and the experimental settings designed to assess within-dataset performance as well as cross-subject, cross-task, and cross-dataset generalization.

All experiments were executed through a version-controlled computational workflow. For every run, the workflow stores the resolved configuration of data preparation, validation and model settings together with the run outputs needed to reconstruct the reported evaluation metrics. These records provide run-level provenance for the results and permit the correspondence between a reported metric and the experiment that produced it to be inspected. The complete implementation, experiment configurations and run-level artifacts are available at `https://github.com/sovetskiysn/eeg-validation-strategies`.

## Dataset alignment and integration rationale

Effective evaluation of cross-task and cross-dataset generalization relies on the availability of datasets that are compatible both technically and conceptually. In this study, we integrate two publicly available EEG datasets that differ in their original purpose and labeling schemes, yet retain sufficient technical and conceptual compatibility for a meaningful comparison.

A major strength of this dataset combination lies in their acquisition setup. Both datasets were recorded using Emotiv EEG systems with the same sampling rate (128 Hz) and voltage resolution, although their original electrode montages differed (14 versus 32 channels). For cross-dataset analyses, only the 12 channels common to both datasets were retained. Differences in EEG recording systems and electrode configurations can hinder cross-dataset transferability [17]. Restricting the analysis to the shared channels substantially reduces this configuration-related mismatch, although it does not eliminate all acquisition-related distribution shifts. Within the limited set of publicly available datasets suitable for this comparison, this pair offered a comparatively compatible basis for the present comparative evaluation.

The conceptual rationale for combining the datasets rests on the cognitive conditions represented in their respective protocols. The mental attention-state dataset contains explicit attention-state labels (*Focused*, *Unfocused*, and *Drowsy*) collected during a low-interaction simulation task, whereas SAM-40 was originally designed for stress assessment. Both capture cognitive conditions strongly tied to attentional processes. The SAM-40 dataset includes tasks widely recognized in cognitive neuroscience for eliciting high levels of attentional engagement:

- **Stroop Color-Word Test**: probes both *selective* and *sustained attention*, requiring participants to suppress automatic reading responses and continually focus on the ink color throughout the task [18–20].
- **Mental arithmetic task**: demands *sustained internal attention* and working memory for continuous manipulation of numerical information [21, 22].
- **Mirror Image Recognition**: engages visuospatial processing and requires both *selective* and *sustained attention* to discriminate between symmetric and asymmetric patterns, relying on mental rotation processes that demand continuous cognitive effort [23–25].

The simulator-based and task-based datasets correspond, respectively, to the *naturalistic stimulus-based protocols* and *continuous cognitive engagement* paradigms described in [16]. In the unified labeling scheme, *Rest* and *Unfocused* are treated as low-attention states, whereas *Focused*, *Stroop*, *Arithmetic*, and *Mirror recognition* are treated as high-attention states. The *Drowsy* phase is excluded from this mapping and from all subsequent preprocessing and analyses because permitted eye closure and drowsiness may confound the comparison with differences in arousal, visual input, and ocular activity. This mapping is summarized in Table 1.

**Table 1. Unified binary attention label mapping across datasets.**

| Attention State | Conditions |
|---|---|
| High Attention | Focused, Stroop, Arithmetic, Mirror |
| Low Attention | Unfocused, Rest |

## Validation strategies

We evaluated the decoders under four validation protocols, each addressing a distinct generalization question. To preserve the independence of the training and test data, all windows originating from the same recording unit were assigned to the same partition. This prevents overlapping windows from being divided between training and test sets, reducing the risk of overly optimistic performance estimates due to temporal autocorrelation between neighboring EEG windows [16]. The within-domain baseline estimates performance when training and test data share the same dataset and condition composition but originate from different recording units. The remaining protocols examine whether discriminative performance is retained when training and testing are separated by participant, task composition, or dataset. These protocols were applied only where the structure of each release supported their interpretation: multiple tasks in SAM-40 enabled cross-task validation. Model fitting was confined to the training data, and the resulting decoder was evaluated on held-out data without target-specific adaptation. Any class imbalance was addressed through inverse-frequency class weighting calculated independently within each training fold, while test data retained their observed class distribution.

**Baseline evaluation.** Reference performance was estimated using five-fold stratified group cross-validation, separately for the mental attention-state dataset, the SAM-40 dataset, and each task-specific composition of SAM-40. The grouping variable was the physical recording unit: one selected session in the mental attention-state dataset and one run-level recording unit in SAM-40, comprising all task recordings included in the analysis for that run. Training and test folds thus shared the dataset and condition composition and could contain data from the same participants, but never from the same recording unit. Keeping each unit intact prevented temporally adjacent or partially overlapping windows from the same acquisition from being assigned to opposite folds, thereby avoiding window-level temporal leakage. These estimates serve as within-domain comparators for the transfer protocols.

**Cross-subject validation.** In each leave-one-subject-out fold, the decoder was trained on all selected data from the remaining participants and tested on all selected data from the held-out participant. The resulting performance quantifies zero-shot generalization to an unseen participant.

**Cross-task validation.** In SAM-40, each task composition represented the same binary contrast of relaxation versus high-demand task conditions, rather than classification of task identity. Cross-task transfer therefore asked whether a decoder fitted to this contrast under one task composition retained discriminative performance under another.

The cross-task folds were also subject-disjoint. For each fold, the source composition from all but one participant formed the training set, whereas the target composition from the held-out participant formed the test set. The protocol thus evaluates the combined shift to an unseen participant and a different task composition, rather than within-participant transfer between tasks. This distinction prevents participant-specific EEG patterns from contributing to apparent cross-task generalization [26, 27].

**Cross-dataset validation.** To evaluate transfer between datasets, models were trained on all available data from a source composition and tested on a composition from the other dataset, in both transfer directions and without target-specific adaptation. SAM-40 was considered as a complete composition and through its task-specific compositions. Unlike the preceding protocols, cross-dataset validation changes several factors simultaneously, including the participant population, experimental paradigm, recording setup, and condition definitions. It therefore measures transfer under their compound distribution shift rather than attributing a performance change to any single factor [17]. The common sampling rate, harmonized binary labels, and shared 12-channel montage establish a compatible input and prediction space, but do not remove the substantive differences between the two datasets.

## Data preparation

The same preparation pipeline was applied to both datasets and across all validation protocols. Its filtering, ICA, channel alignment and windowing parameters were fixed in advance and were not tuned separately for a dataset, task composition, validation protocol or decoder. The purpose was to place recordings in a common input space while limiting preprocessing itself as an additional source of variation. Differences between validation results are therefore interpreted against a fixed preparation procedure. The pipeline was instantiated as a fixed set of dataset preparations, that is, the analytical compositions on which every validation protocol was run and are collected in Table 2. The remainder of this subsection describes the selection, cleaning, alignment and segmentation steps that produced them.

**Table 2. Analytical dataset compositions.**

| Dataset configuration | Retained conditions | Participants | Recording units | Windows, n (low / high) | Class distribution, % (low / high) |
|---|---|---|---|---|---|
| Full Dataset A | Focused, Unfocused | 5 | 24 | 3,192 / 3,192 | 50.0 / 50.0 |
| Full Dataset B | Rest, Stroop, Arithmetic, Mirror | 40 | 120 | 600 / 1,800 | 25.0 / 75.0 |
| B: Stroop | Rest, Stroop | 40 | 120 | 600 / 600 | 50.0 / 50.0 |
| B: Arithmetic | Rest, Arithmetic | 40 | 120 | 600 / 600 | 50.0 / 50.0 |
| B: Mirror | Rest, Mirror | 40 | 120 | 600 / 600 | 50.0 / 50.0 |

**Recording and label selection.** Recordings were selected before signal processing. Following the source study, the first two recording sessions for each participant in the mental attention-state dataset were intended for habituation to the simulator and were not included in the analyses [14]. The *Drowsy* condition was excluded from the binary classification analysis because participants were permitted to close their eyes and enter a drowsy state. This condition differs from the *Focused* and *Unfocused* conditions not only in protocol-defined task demand, but also in arousal, visual input and ocular activity. Including it in the low-attention class would therefore confound the intended contrast. The drowsy segment was removed before filtering and ICA, so no drowsy epochs were generated or presented to a classifier. In contrast, the complete collection of SAM-40 recordings was retained for analysis. SAM-40 provides the task recordings of each trial as separate files. Within each scenario, the task recordings selected from one trial were grouped into a single recording unit, whereas task recordings not selected by the scenario were not carried forward to later preparation steps. Accordingly, the binary mapping in Table 1 was applied to the retained recordings from the two datasets.

**Filtering and ICA cleaning.** Filtering was applied to each recording, and ICA was fitted within participant-specific acquisition groupings. For the mental attention-state dataset, each grouping corresponded to a selected session, whereas for SAM-40 it comprised the selected task recordings of a participant. To prevent preprocessing from becoming an additional source of variation, we used a deliberately restrained, conventional procedure: standard filtering followed by ICA-based artifact cleaning. Keeping preparation deliberately simple allows differences in performance to be interpreted primarily in relation to the recordings and validation protocols, rather than to a complex preparation pipeline. EEG channels were notch-filtered at 50 Hz, band-pass filtered from 1 to 45 Hz and rereferenced to the average reference. Artifact correction used independent component analysis with extended-Infomax optimization, retaining components that explained 99% of the variance. The resulting components were automatically classified from their spatial, spectral and temporal characteristics. Components identified as eye-blink or muscle artifacts with a probability of at least 0.80 were removed. The ICA random state was fixed by the experiment-wide configuration.

**Channel alignment and window segmentation.** After signal cleaning, each recording was restricted to the 12 channels shared by the two datasets: F3, F4, F7, F8, FC5, FC6, O1, O2, P7, P8, T7 and T8. Each retained labelled block was segmented into 5-s windows with 0.5-s overlap. Every retained window inherited the label of its source block and metadata identifying its dataset, participant, session, task, run, recording unit and temporal onset. The recording-unit identifier was used by the validation protocols to ensure that all overlapping windows from a unit remained on the same side of a split.

The fixed selection, cleaning, alignment and segmentation steps therefore yielded a deterministic analytical window set for each dataset composition. All model-specific input representations were derived from these same prepared windows, rather than from separately prepared data. The retained recording units, window counts and observed class distributions of the resulting compositions are those reported in Table 2. Class weights were calculated independently within each training fold. All Dataset B configurations contain the same 40 participants and differ only in the task recordings retained alongside the resting-state recordings.

## Models

We evaluated five classifiers rather than relying on a single decoder to test whether the conclusions are robust to a change in modelling assumptions. This makes the decoder an explicit factor of the design: a difference between validation protocols that is reproduced across models is more credibly attributed to the data and evaluation protocol than to the idiosyncrasies of one learner. The selected models span three families used in EEG decoding: linear, tree-based, and deep-learning architectures. The deep networks are included because they learn task-relevant representations directly from the channel-by-time EEG windows, reducing reliance on a manually specified feature set. All models were trained to predict whether a window belonged to a high-demand or a low-demand condition in the sense of Table 1

Hyperparameters were fixed *a priori* and held constant across datasets, task compositions, validation protocols and runs. No hyperparameter search was performed. Values followed library defaults or published reference recipes, except for the departures specified below. A fixed random seed governed fold construction, model initialization and network training. The resolved configuration of each run was stored with its outputs.

**Traditional machine learning.** Both classical classifiers used the same fixed, label-free feature representation. For each of the 12 channels, we computed log relative power in the 1–4 Hz, 4–8 Hz, 8–13 Hz and 13–45 Hz bands, together with sample entropy, mean, variance, skewness and kurtosis. Relative band power was normalized by the channel total before log transformation. The nine features from each channel formed one 108-dimensional feature vector per window.

Logistic regression provides a transparent linear baseline. It estimates a linear decision boundary from the feature vectors, allowing the contribution of individual features to be inspected. Its feature vectors were z-score standardized using the mean and standard deviation estimated from the training fold only.

XGBoost [28] represents tree-based ensemble learning, allowing nonlinear relations and feature interactions to be modelled within the same feature representation. It was also the decoder used in the originally submitted version of this study, preserving a direct point of comparison with those results. It received the same feature vectors without scaling. For both classical models, inverse-frequency class weights were calculated from the training fold only.

**Deep learning.** The remaining three classifiers operate directly on the prepared channel-by-time EEG windows, rather than on the classical feature vectors. Each input window had a $12 \times 640$ channel-by-time shape and was standardized using statistics estimated from the training fold only.

*ShallowFBCSPNet* [29] is a shallow convolutional neural network designed to emulate the main operations of the filter-bank common spatial pattern (FBCSP) pipeline within a trainable model. It first applies temporal convolution and spatial filtering across channels, then uses a squaring nonlinearity, mean pooling and a logarithmic activation to form band-power-like features. Thus, its inductive bias reflects a conventional EEG decoding approach, while its filters are optimized jointly with the classifier.

*EEGNet* [30] is a compact convolutional neural network designed specifically for EEG. Its temporal convolutions learn filters over time, depthwise convolutions learn spatial filters across electrodes for each temporal feature map, and separable convolutions combine these learned features efficiently. This factorized design substantially limits the number of trainable parameters while retaining separate temporal and spatial feature learning.

*EEG Conformer* [31] combines a convolutional front end with a Transformer encoder. The convolutional module forms local temporal and spatial features and converts them into a sequence of embeddings. Self-attention in the Transformer then models global dependencies between these embeddings. It therefore complements the local receptive fields of convolution with a mechanism for relating EEG patterns over longer time spans. To limit the risk of overfitting in the smallest group-disjoint training folds, we reduced the capacity of the Braindecode default configuration before evaluation and then fixed it across all scenarios. Specifically, we reduced the number of temporal convolution filters, which also determines the embedding width, from 40 to 16, the number of Transformer encoder layers from 6 to 2, and the number of attention heads from 10 to 4. This retained the convolution–attention architecture while reducing its parameter count for the $12 \times 640$ input windows from 527,746 to 170,466.

All three networks were trained from scratch, with no pretrained weights, for 100 epochs with a batch size of 64. Their loss functions used inverse-frequency class weights calculated from the training fold only, and no resampling was applied.

## Results and analysis

To characterize zero-shot generalization across the prespecified validation strategies, we evaluated the binary contrast between high-demand conditions and rest or task disengagement with five decoders. Table 3 reports decoder-specific performance across these strategies. It presents participant-level balanced accuracy, macro F1, MCC, ROC-AUC, and per-class recall for logistic regression, the interpretable reference decoder. Corresponding tables for XGBoost, EEGNet, ShallowFBCSPNet, and EEGConformer are provided in the Supporting Information.

**Table 3. Performance metrics across experimental scenarios for Logistic Regression.**

| Scenario | Validation method | Balanced Accuracy | Macro F1 | MCC | ROC AUC | Per-class Recall (Low/High) |
|---|---|---|---|---|---|---|
| Baseline (within) | Stratified K-fold(Full A) | 0.64 | 0.63 | 0.30 | 0.73 | 0.62/0.66 |
| | Stratified K-fold(Full B) | 0.68 | 0.64 | 0.34 | 0.74 | 0.65/0.71 |
| | Stratified K-fold(B: Stroop) | 0.69 | 0.68 | 0.40 | 0.74 | 0.67/0.71 |
| | Stratified K-fold(B: Arithmetic) | 0.66 | 0.65 | 0.35 | 0.74 | 0.66/0.67 |
| | Stratified K-fold(B: Mirror) | 0.69 | 0.67 | 0.39 | 0.75 | 0.66/0.72 |
| Cross-subject | Leave-one-subject-out(Full A) | 0.63 | 0.61 | 0.29 | 0.73 | 0.63/0.63 |
| | Leave-one-subject-out(Full B) | 0.66 | 0.61 | 0.31 | 0.74 | 0.63/0.69 |
| Cross-task (Dataset B) | Full B → Stroop | 0.66 | 0.64 | 0.34 | 0.73 | 0.64/0.68 |
| | Full B → Arithmetic | 0.64 | 0.62 | 0.31 | 0.72 | 0.64/0.65 |
| | Full B → Mirror | 0.67 | 0.65 | 0.36 | 0.77 | 0.63/0.71 |
| | Stroop → Full B | 0.63 | 0.58 | 0.25 | 0.67 | 0.63/0.63 |
| | Stroop → Arithmetic | 0.58 | 0.55 | 0.16 | 0.60 | 0.65/0.50 |
| | Stroop → Mirror | 0.63 | 0.60 | 0.27 | 0.68 | 0.64/0.62 |
| | Arithmetic → Full B | 0.62 | 0.55 | 0.23 | 0.68 | 0.66/0.58 |
| | Arithmetic → Stroop | 0.56 | 0.53 | 0.14 | 0.61 | 0.67/0.46 |
| | Arithmetic → Mirror | 0.66 | 0.64 | 0.35 | 0.73 | 0.67/0.66 |
| | Mirror → Full B | 0.64 | 0.57 | 0.26 | 0.70 | 0.68/0.60 |
| | Mirror → Stroop | 0.60 | 0.57 | 0.21 | 0.65 | 0.66/0.54 |
| | Mirror → Arithmetic | 0.62 | 0.59 | 0.27 | 0.70 | 0.63/0.62 |
| Cross-dataset | Full A → Full B | 0.52 | 0.37 | 0.03 | 0.54 | 0.78/0.26 |
| | Full B → Full A | 0.55 | 0.53 | 0.12 | 0.58 | 0.35/0.76 |
| | Full A → Stroop | 0.49 | 0.42 | -0.03 | 0.51 | 0.71/0.27 |
| | Stroop → Full A | 0.54 | 0.53 | 0.08 | 0.56 | 0.45/0.63 |
| | Full A → Arithmetic | 0.53 | 0.48 | 0.08 | 0.58 | 0.68/0.38 |
| | Arithmetic → Full A | 0.52 | 0.52 | 0.04 | 0.54 | 0.43/0.61 |
| | Full A → Mirror | 0.53 | 0.47 | 0.06 | 0.55 | 0.72/0.34 |
| | Mirror → Full A | 0.52 | 0.48 | 0.06 | 0.54 | 0.26/0.79 |

## Generalization under increasingly stringent validation strategies

To examine generalization under increasingly stringent validation strategies, we compared directed cross-task and cross-dataset transfer with the baseline scenario for each target composition. These comparisons place performance under progressively less similar training and test data in relation to a target-specific reference. They therefore assess whether the binary contrast was retained without target-specific adaptation as the participant, task composition, or dataset differed between training and test data.

Figure 1 summarizes the directed cross-task and cross-dataset comparisons. Rows are grouped by source composition and decoder, and columns denote target compositions. Each nonmissing matrix cell reports balanced accuracy calculated separately from the held-out windows of each target participant and then averaged across target participants for the given decoder and direction. Off-diagonal cells represent zero-shot transfers without target-specific adaptation, whereas the grey diagonal reports the target-specific baseline estimated with five-fold stratified group validation that keeps recording units disjoint between training and test folds. For each target represented in a summary column, we first averaged participant-level estimates across the five decoders. We then averaged these decoder-level transfer and baseline values across targets and calculated their difference. Negative values therefore indicate lower transfer performance than the applicable target baselines, whereas dashes identify comparisons that are not applicable.

Figure 2 compares baseline and leave-one-subject-out cross-subject validation for the mental attention-state dataset and the SAM-40 dataset. For each decoder and scenario, balanced accuracy was calculated separately from the held-out windows of each target participant and then averaged across target participants, and the coloured lines show these decoder-specific means. Black points show the mean of the five decoder-specific estimates. Their bars show paired 95% percentile bootstrap intervals from 10,000 target-participant resamples, with the same resample used for all decoders to preserve their pairing.

| | | Target (test) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Cross-task | | | | Cross-dataset | | | | | | |
| Source (train) | | Stroop | Arithmetic | Mirror | Full B | Stroop | Arithmetic | Mirror | Full B | Full A | Baseline vs cross-task mean diff | Baseline vs cross-dataset mean diff |
| Stroop | Logistic reg | 69.3 | 57.7 | 62.7 | 63.1 | 69.3 | – | – | – | 54.0 | -9.6 | -13.1 |
| | XGBoost | 69.0 | 55.7 | 61.9 | 60.8 | 69.0 | – | – | – | 50.6 | | |
| | EEGNet | 66.1 | 53.8 | 58.8 | 59.4 | 66.1 | – | – | – | 50.3 | | |
| | ShallowNet | 78.5 | 59.3 | 64.1 | 65.4 | 78.5 | – | – | – | 55.5 | | |
| | EEGConformer | 73.2 | 55.6 | 61.2 | 61.8 | 73.2 | – | – | – | 52.2 | | |
| Arithmetic | Logistic reg | 56.0 | 66.4 | 66.2 | 62.2 | – | 66.4 | – | – | 52.2 | -6.3 | -10.4 |
| | XGBoost | 58.2 | 69.5 | 66.6 | 63.6 | – | 69.5 | – | – | 58.9 | | |
| | EEGNet | 58.9 | 65.3 | 64.3 | 61.9 | – | 65.3 | – | – | 50.3 | | |
| | ShallowNet | 62.9 | 76.2 | 71.0 | 68.1 | – | 76.2 | – | – | 55.6 | | |
| | EEGConformer | 61.8 | 71.2 | 69.4 | 67.2 | – | 71.2 | – | – | 59.5 | | |
| Mirror | Logistic reg | 59.7 | 62.3 | 68.6 | 63.8 | – | – | 68.6 | – | 52.4 | -6.6 | -13.1 |
| | XGBoost | 61.5 | 64.3 | 70.5 | 64.9 | – | – | 70.5 | – | 53.2 | | |
| | EEGNet | 61.2 | 63.3 | 67.0 | 64.1 | – | – | 67.0 | – | 48.8 | | |
| | ShallowNet | 63.2 | 67.4 | 73.9 | 67.5 | – | – | 73.9 | – | 54.9 | | |
| | EEGConformer | 60.9 | 64.6 | 69.1 | 64.6 | – | – | 69.1 | – | 53.6 | | |
| Full B | Logistic reg | 66.2 | 64.5 | 67.1 | 68.3 | – | – | – | 68.3 | 55.2 | -3.1 | -11.0 |
| | XGBoost | 64.1 | 65.1 | 67.2 | 68.0 | – | – | – | 68.0 | 54.1 | | |
| | EEGNet | 65.8 | 64.6 | 68.4 | 66.9 | – | – | – | 66.9 | 51.0 | | |
| | ShallowNet | 66.9 | 67.7 | 67.0 | 71.9 | – | – | – | 71.9 | 54.8 | | |
| | EEGConformer | 70.7 | 70.2 | 72.6 | 71.8 | – | – | – | 71.8 | 58.2 | | |
| Full A | Logistic reg | – | – | – | – | 48.8 | 53.4 | 53.2 | 52.1 | 64.0 | – | -20.9 |
| | XGBoost | – | – | – | – | 47.8 | 54.0 | 51.4 | 51.6 | 67.0 | | |
| | EEGNet | – | – | – | – | 44.7 | 49.2 | 45.7 | 46.7 | 66.6 | | |
| | ShallowNet | – | – | – | – | 47.5 | 51.7 | 47.7 | 49.2 | 66.4 | | |
| | EEGConformer | – | – | – | – | 42.2 | 52.5 | 47.3 | 47.0 | 64.3 | | |

0 25 50 75 100
Balanced accuracy, %

**Fig 1. Zero-shot transfer performance across source and target compositions.** Rows and columns denote source and target compositions. The grey diagonal shows the target-specific baseline scenario. The two right-hand columns show mean differences from the relevant target baselines. Negative values indicate lower performance, and dashes mark contrasts that are not applicable.

For cross-task transfer, the mean difference from the target-specific baseline, averaged across all five decoders, was 9.6, 6.3, 6.6, and 3.1 percentage points lower for the Stroop, Arithmetic, and Mirror task compositions and the SAM-40 dataset, respectively. The mean decrease across these sources was 6.4 percentage points. For cross-dataset transfer, the decoder-averaged decreases for the Stroop, Arithmetic, and Mirror task compositions, the SAM-40 dataset, and the mental attention-state dataset were 13.1, 10.4, 13.1, 11.0, and 20.9 percentage points, respectively. The mean decrease was 13.7 percentage points. Thus, zero-shot transfer reduced performance in every eligible comparison, with a larger reduction when training and test data came from different datasets.

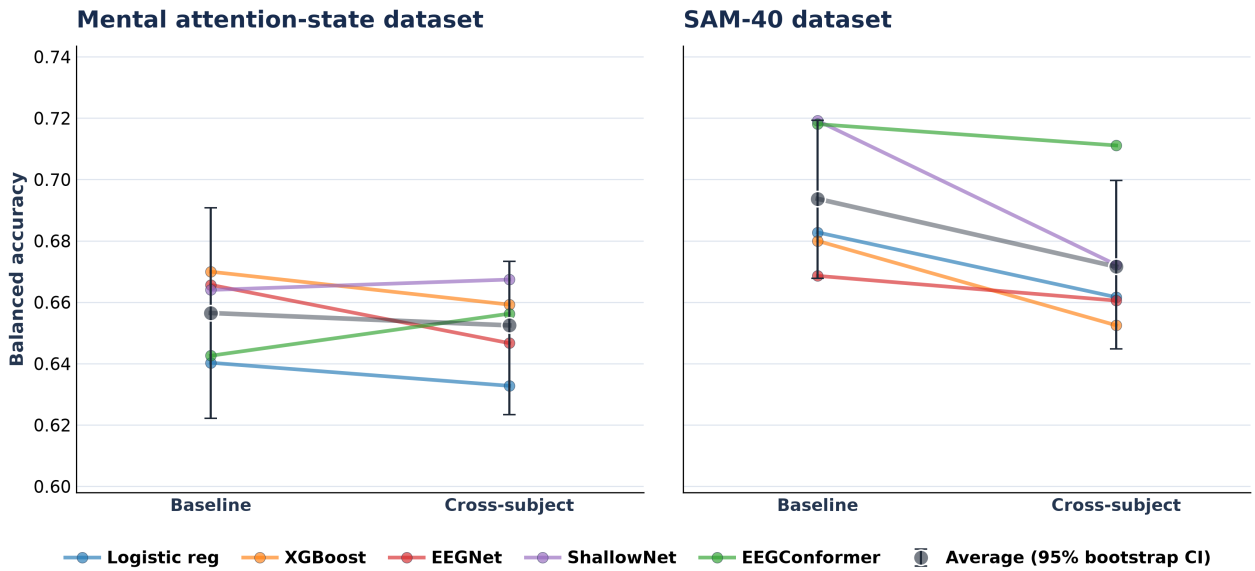


**Fig 2. Baseline and cross-subject performance across decoders.**

In the mental attention-state dataset, mean balanced accuracy was 65.5% in the baseline scenario and 65.4% in cross-subject validation, a decrease of 0.1 percentage points. In the SAM-40 dataset, it decreased from 69.1% to 67.1%, a difference of 2.0 percentage points.

With two exceptions, all decoder-specific estimates fell within the corresponding 95% bootstrap interval. In the SAM-40 baseline scenario, ShallowFBCSPNet reached 72.0%, which was slightly above the interval upper bound of 71.6%. In cross-subject validation, EEGConformer reached 70.7%, which was above the upper bound of 69.7%.

## Performance consistency across decoders

To assess whether performance patterns across validation strategies depended on decoder choice, Figure 3 compares all validation scenarios across the five decoders. Each coloured point represents the balanced accuracy of one decoder, averaged across target participants in a given scenario. The black point and horizontal bar show the mean across the five decoders and its paired 95% percentile bootstrap confidence interval (CI), respectively, estimated from 10,000 target-participant resamples. The same resample was used for all decoders to preserve their pairing.

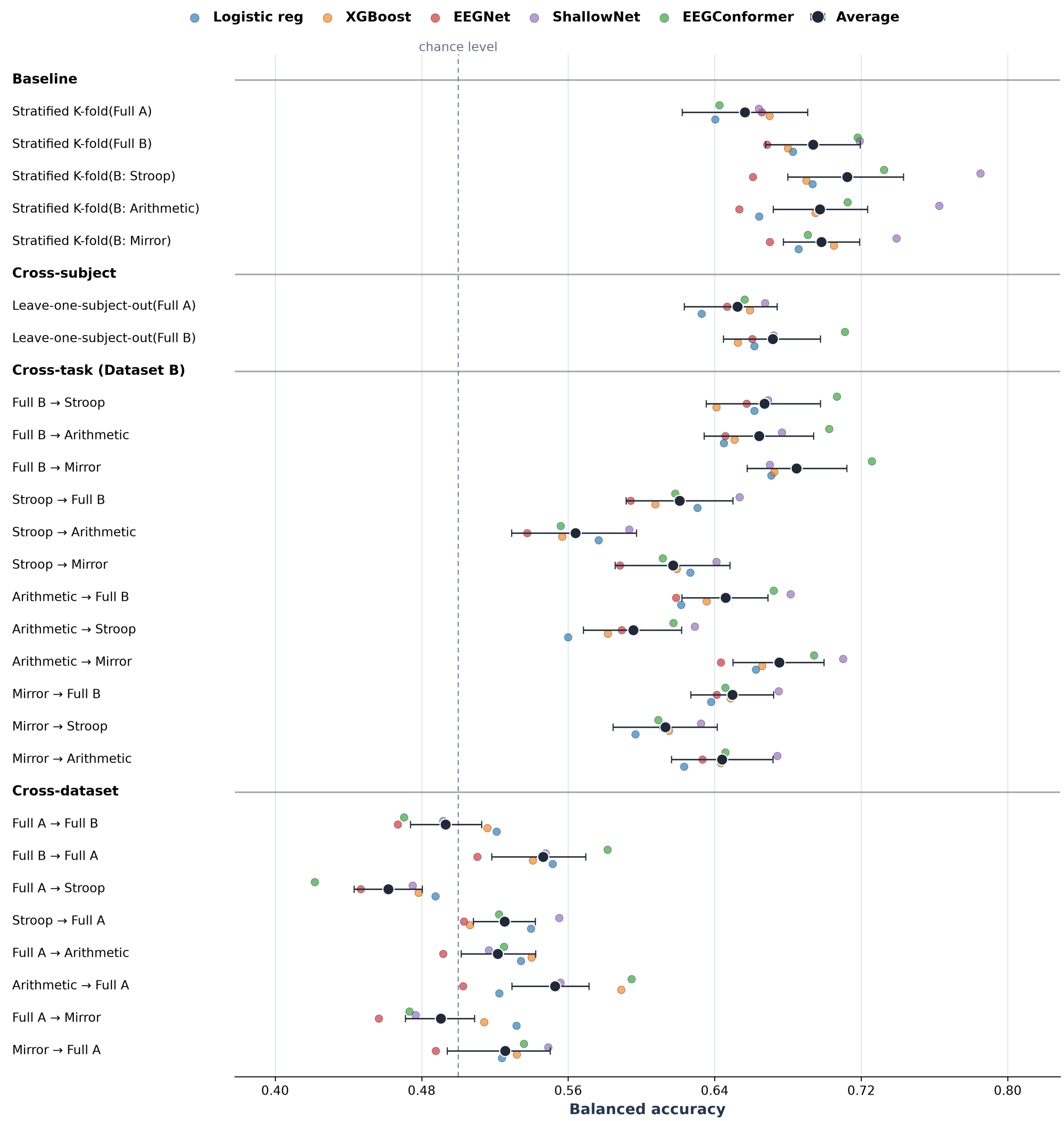


**Fig 3. Participant-level balanced accuracy across experimental scenarios for the five decoders.**

Across the 27 scenarios, 42 of 135 coloured decoder points lay outside the corresponding 95% bootstrap interval. Logistic Regression lay outside the interval in seven scenarios, with four points below and three above the interval. XGBoost lay outside the interval in four scenarios, with one point below and three above the interval. All of these departures occurred in cross-dataset transfer settings.

EEGNet lay outside the interval in 12 scenarios, always below it. ShallowFBCSPNet lay outside the interval in 10 scenarios, always above it. These were the two decoders with the most frequent departures from the scenario-average interval.

EEGConformer lay outside the interval in nine scenarios, with two points below and seven above the interval. Overall, these patterns show that decoder choice materially influenced balanced accuracy across validation scenarios, including both within-dataset baselines and transfer settings.

# Discussion

Across the data compositions and validation protocols examined here, zero-shot discriminative performance declined as the difference between the training and test data increased. Performance was comparatively stable when generalization was limited to an unseen participant within a dataset, but was lower when the training and test compositions differed by task and lower still when they came from different datasets. Thus, for these protocol-defined high- versus low-demand contrasts, technical compatibility and a common binary label space were insufficient to preserve target-specific performance under increasingly heterogeneous zero-shot transfer conditions.

The overall degradation pattern was observed across the five decoders, but the absolute level of balanced accuracy varied materially with decoder choice. This finding makes the decoder an important factor in comparative EEG evaluation rather than an interchangeable implementation detail. In particular, conclusions about a validation scenario should distinguish a pattern that is reproduced across model families from the performance of any single predeclared model recipe.

These results provide leakage-resistant, pre-adaptation baselines for assessing source–target suitability in the studied data compositions. They indicate that nominally aligned labels and partially harmonized acquisition specifications do not by themselves establish that a source dataset will support useful zero-shot transfer to a target dataset or task composition. Such baselines can therefore inform source selection, which should be based on empirical, directed source–target comparisons rather than on nominal label alignment alone. By relating each transfer result to the corresponding target-specific baseline, the analysis distinguishes preservation of the contrast under transfer from the discriminability of that contrast in the target data itself. The resulting references also make it possible to evaluate whether subsequent transfer-learning, domain-adaptation, or personalization methods provide a meaningful improvement over matched zero-shot performance rather than merely achieving a favourable absolute score.

Several limitations delimit the interpretation of these findings. First, the binary labels represent protocol-defined contrasts and do not isolate attention from related differences in arousal, visual input, stress, or task demands. Second, selection of only two technically comparable datasets enabled a common input space for cross-dataset evaluation, but limited the volume and diversity of data available for model fitting. The analysis also used the 12 shared channels, a fixed parsimonious preparation pipeline, and predeclared model recipes without hyperparameter or architecture selection to provide comparable pre-adaptation benchmarks rather than to maximize performance in each scenario. The reported figures consequently do not represent the maximum attainable performance for an individual scenario.

Future work could usefully optimize hyperparameters and architectural choices within training folds, which would help distinguish limitations of the predeclared recipes from those imposed by the data compositions and transfer scenarios. It would also be informative to assess whether domain-adaptation methods offer value relative to the same models without adaptation. Such comparisons could provide a clearer indication of the extent to which adaptation improves transfer-learning performance beyond the pre-adaptation benchmarks reported here. To preserve the interpretability of these estimates, future comparisons would benefit from retaining group-disjoint, leakage-resistant splits and from avoiding the use of target test data for model selection or parameter tuning.

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## Supporting information

**Table 4. Performance metrics across experimental scenarios for XGBoost.**

| Scenario | Validation method | Balanced Accuracy | Macro F1 | MCC | ROC AUC | Per-class Recall (Low/High) |
|---|---|---|---|---|---|---|
| Baseline (within) | Stratified K-fold(Full A) | 0.67 | 0.67 | 0.35 | 0.74 | 0.64/0.70 |
| | Stratified K-fold(Full B) | 0.68 | 0.65 | 0.36 | 0.75 | 0.60/0.76 |
| | Stratified K-fold(B: Stroop) | 0.69 | 0.68 | 0.40 | 0.76 | 0.68/0.70 |
| | Stratified K-fold(B: Arithmetic) | 0.70 | 0.68 | 0.41 | 0.76 | 0.69/0.70 |
| | Stratified K-fold(B: Mirror) | 0.70 | 0.70 | 0.43 | 0.77 | 0.72/0.69 |
| Cross-subject | Leave-one-subject-out(Full A) | 0.66 | 0.65 | 0.33 | 0.74 | 0.65/0.67 |
| | Leave-one-subject-out(Full B) | 0.65 | 0.61 | 0.31 | 0.73 | 0.58/0.72 |
| Cross-task (Dataset B) | Full B → Stroop | 0.64 | 0.61 | 0.31 | 0.71 | 0.60/0.68 |
| | Full B → Arithmetic | 0.65 | 0.63 | 0.33 | 0.71 | 0.59/0.71 |
| | Full B → Mirror | 0.67 | 0.65 | 0.37 | 0.77 | 0.58/0.77 |
| | Stroop → Full B | 0.61 | 0.54 | 0.22 | 0.67 | 0.63/0.58 |
| | Stroop → Arithmetic | 0.56 | 0.52 | 0.13 | 0.58 | 0.67/0.45 |
| | Stroop → Mirror | 0.62 | 0.59 | 0.27 | 0.69 | 0.65/0.59 |
| | Arithmetic → Full B | 0.64 | 0.56 | 0.25 | 0.70 | 0.68/0.59 |
| | Arithmetic → Stroop | 0.58 | 0.55 | 0.18 | 0.62 | 0.68/0.48 |
| | Arithmetic → Mirror | 0.67 | 0.64 | 0.36 | 0.74 | 0.66/0.67 |
| | Mirror → Full B | 0.65 | 0.58 | 0.28 | 0.73 | 0.71/0.59 |
| | Mirror → Stroop | 0.61 | 0.58 | 0.25 | 0.67 | 0.72/0.51 |
| | Mirror → Arithmetic | 0.64 | 0.62 | 0.32 | 0.72 | 0.70/0.59 |
| Cross-dataset | Full A → Full B | 0.52 | 0.49 | 0.04 | 0.52 | 0.28/0.75 |
| | Full B → Full A | 0.54 | 0.52 | 0.09 | 0.55 | 0.37/0.71 |
| | Full A → Stroop | 0.48 | 0.42 | -0.04 | 0.47 | 0.26/0.70 |
| | Stroop → Full A | 0.51 | 0.48 | 0.02 | 0.50 | 0.32/0.69 |
| | Full A → Arithmetic | 0.54 | 0.47 | 0.10 | 0.55 | 0.27/0.81 |
| | Arithmetic → Full A | 0.59 | 0.58 | 0.18 | 0.62 | 0.63/0.55 |
| | Full A → Mirror | 0.51 | 0.45 | 0.05 | 0.52 | 0.27/0.76 |
| | Mirror → Full A | 0.53 | 0.52 | 0.07 | 0.56 | 0.61/0.45 |

**Table 5. Performance metrics across experimental scenarios for EEGNet.**

| Scenario | Validation method | Balanced Accuracy | Macro F1 | MCC | ROC AUC | Per-class Recall (Low/High) |
|---|---|---|---|---|---|---|
| Baseline (within) | Stratified K-fold(Full A) | 0.67 | 0.66 | 0.34 | 0.74 | 0.57/0.76 |
| | Stratified K-fold(Full B) | 0.67 | 0.64 | 0.33 | 0.74 | 0.58/0.76 |
| | Stratified K-fold(B: Stroop) | 0.66 | 0.64 | 0.35 | 0.73 | 0.59/0.73 |
| | Stratified K-fold(B: Arithmetic) | 0.65 | 0.64 | 0.33 | 0.71 | 0.63/0.68 |
| | Stratified K-fold(B: Mirror) | 0.67 | 0.66 | 0.36 | 0.74 | 0.62/0.72 |
| Cross-subject | Leave-one-subject-out(Full A) | 0.65 | 0.63 | 0.32 | 0.76 | 0.62/0.68 |
| | Leave-one-subject-out(Full B) | 0.66 | 0.62 | 0.33 | 0.75 | 0.59/0.73 |
| Cross-task (Dataset B) | Full B → Stroop | 0.66 | 0.63 | 0.35 | 0.74 | 0.61/0.70 |
| | Full B → Arithmetic | 0.65 | 0.62 | 0.33 | 0.73 | 0.59/0.70 |
| | Full B → Mirror | 0.68 | 0.67 | 0.40 | 0.76 | 0.60/0.77 |
| | Stroop → Full B | 0.59 | 0.53 | 0.19 | 0.67 | 0.57/0.62 |
| | Stroop → Arithmetic | 0.54 | 0.50 | 0.08 | 0.59 | 0.62/0.45 |
| | Stroop → Mirror | 0.59 | 0.55 | 0.19 | 0.66 | 0.59/0.59 |
| | Arithmetic → Full B | 0.62 | 0.56 | 0.24 | 0.69 | 0.61/0.63 |
| | Arithmetic → Stroop | 0.59 | 0.55 | 0.20 | 0.64 | 0.62/0.56 |
| | Arithmetic → Mirror | 0.64 | 0.62 | 0.32 | 0.70 | 0.63/0.66 |
| | Mirror → Full B | 0.64 | 0.60 | 0.27 | 0.69 | 0.62/0.66 |
| | Mirror → Stroop | 0.61 | 0.59 | 0.24 | 0.65 | 0.61/0.61 |
| | Mirror → Arithmetic | 0.63 | 0.61 | 0.30 | 0.68 | 0.61/0.66 |
| Cross-dataset | Full A → Full B | 0.47 | 0.34 | -0.07 | 0.46 | 0.61/0.32 |
| | Full B → Full A | 0.51 | 0.48 | 0.04 | 0.53 | 0.30/0.72 |
| | Full A → Stroop | 0.45 | 0.37 | -0.13 | 0.39 | 0.60/0.29 |
| | Stroop → Full A | 0.50 | 0.46 | 0.01 | 0.50 | 0.27/0.74 |
| | Full A → Arithmetic | 0.49 | 0.43 | -0.00 | 0.52 | 0.58/0.40 |
| | Arithmetic → Full A | 0.50 | 0.43 | 0.01 | 0.51 | 0.16/0.84 |
| | Full A → Mirror | 0.46 | 0.38 | -0.10 | 0.46 | 0.60/0.32 |
| | Mirror → Full A | 0.49 | 0.48 | -0.02 | 0.48 | 0.44/0.53 |

**Table 6. Performance metrics across experimental scenarios for ShallowFBCSPNet.**

| Scenario | Validation method | Balanced Accuracy | Macro F1 | MCC | ROC AUC | Per-class Recall (Low/High) |
|---|---|---|---|---|---|---|
| Baseline (within) | Stratified K-fold(Full A) | 0.66 | 0.66 | 0.34 | 0.74 | 0.64/0.68 |
| | Stratified K-fold(Full B) | 0.72 | 0.72 | 0.46 | 0.82 | 0.57/0.87 |
| | Stratified K-fold(B: Stroop) | 0.78 | 0.78 | 0.59 | 0.86 | 0.77/0.80 |
| | Stratified K-fold(B: Arithmetic) | 0.76 | 0.76 | 0.54 | 0.84 | 0.76/0.76 |
| | Stratified K-fold(B: Mirror) | 0.74 | 0.73 | 0.49 | 0.81 | 0.74/0.74 |
| Cross-subject | Leave-one-subject-out(Full A) | 0.67 | 0.66 | 0.36 | 0.77 | 0.71/0.63 |
| | Leave-one-subject-out(Full B) | 0.67 | 0.64 | 0.38 | 0.81 | 0.54/0.80 |
| Cross-task (Dataset B) | Full B → Stroop | 0.67 | 0.63 | 0.38 | 0.80 | 0.55/0.79 |
| | Full B → Arithmetic | 0.68 | 0.64 | 0.40 | 0.80 | 0.54/0.81 |
| | Full B → Mirror | 0.67 | 0.64 | 0.37 | 0.80 | 0.56/0.78 |
| | Stroop → Full B | 0.65 | 0.59 | 0.29 | 0.75 | 0.68/0.63 |
| | Stroop → Arithmetic | 0.59 | 0.55 | 0.21 | 0.67 | 0.73/0.45 |
| | Stroop → Mirror | 0.64 | 0.61 | 0.31 | 0.72 | 0.74/0.54 |
| | Arithmetic → Full B | 0.68 | 0.61 | 0.35 | 0.79 | 0.74/0.62 |
| | Arithmetic → Stroop | 0.63 | 0.59 | 0.28 | 0.72 | 0.74/0.52 |
| | Arithmetic → Mirror | 0.71 | 0.69 | 0.45 | 0.80 | 0.76/0.66 |
| | Mirror → Full B | 0.68 | 0.62 | 0.33 | 0.78 | 0.69/0.66 |
| | Mirror → Stroop | 0.63 | 0.61 | 0.28 | 0.70 | 0.70/0.57 |
| | Mirror → Arithmetic | 0.67 | 0.65 | 0.37 | 0.77 | 0.67/0.68 |
| Cross-dataset | Full A → Full B | 0.49 | 0.28 | -0.05 | 0.49 | 0.83/0.16 |
| | Full B → Full A | 0.55 | 0.48 | 0.11 | 0.58 | 0.28/0.81 |
| | Full A → Stroop | 0.47 | 0.36 | -0.08 | 0.41 | 0.82/0.13 |
| | Stroop → Full A | 0.56 | 0.52 | 0.12 | 0.60 | 0.52/0.59 |
| | Full A → Arithmetic | 0.52 | 0.42 | 0.04 | 0.57 | 0.79/0.24 |
| | Arithmetic → Full A | 0.56 | 0.53 | 0.14 | 0.61 | 0.55/0.56 |
| | Full A → Mirror | 0.48 | 0.37 | -0.08 | 0.49 | 0.82/0.13 |
| | Mirror → Full A | 0.55 | 0.53 | 0.11 | 0.58 | 0.53/0.57 |

**Table 7. Performance metrics across experimental scenarios for EEGConformer.**

| Scenario | Validation method | Balanced Accuracy | Macro F1 | MCC | ROC AUC | Per-class Recall (Low/High) |
|---|---|---|---|---|---|---|
| Baseline (within) | Stratified K-fold(Full A) | 0.64 | 0.64 | 0.29 | 0.73 | 0.61/0.67 |
| | Stratified K-fold(Full B) | 0.72 | 0.71 | 0.44 | 0.81 | 0.60/0.84 |
| | Stratified K-fold(B: Stroop) | 0.73 | 0.72 | 0.48 | 0.81 | 0.64/0.82 |
| | Stratified K-fold(B: Arithmetic) | 0.71 | 0.69 | 0.46 | 0.81 | 0.58/0.84 |
| | Stratified K-fold(B: Mirror) | 0.69 | 0.67 | 0.41 | 0.79 | 0.57/0.81 |
| Cross-subject | Leave-one-subject-out(Full A) | 0.66 | 0.65 | 0.33 | 0.75 | 0.65/0.66 |
| | Leave-one-subject-out(Full B) | 0.71 | 0.68 | 0.42 | 0.82 | 0.61/0.81 |
| Cross-task (Dataset B) | Full B → Stroop | 0.71 | 0.68 | 0.44 | 0.80 | 0.61/0.80 |
| | Full B → Arithmetic | 0.70 | 0.68 | 0.44 | 0.81 | 0.60/0.80 |
| | Full B → Mirror | 0.73 | 0.71 | 0.48 | 0.82 | 0.63/0.82 |
| | Stroop → Full B | 0.62 | 0.56 | 0.24 | 0.72 | 0.59/0.64 |
| | Stroop → Arithmetic | 0.56 | 0.51 | 0.12 | 0.61 | 0.66/0.45 |
| | Stroop → Mirror | 0.61 | 0.58 | 0.25 | 0.71 | 0.62/0.60 |
| | Arithmetic → Full B | 0.67 | 0.63 | 0.33 | 0.77 | 0.64/0.71 |
| | Arithmetic → Stroop | 0.62 | 0.59 | 0.26 | 0.69 | 0.61/0.62 |
| | Arithmetic → Mirror | 0.69 | 0.67 | 0.41 | 0.79 | 0.63/0.76 |
| | Mirror → Full B | 0.65 | 0.60 | 0.28 | 0.75 | 0.59/0.71 |
| | Mirror → Stroop | 0.61 | 0.58 | 0.23 | 0.68 | 0.58/0.63 |
| | Mirror → Arithmetic | 0.65 | 0.61 | 0.32 | 0.75 | 0.57/0.72 |
| Cross-dataset | Full A → Full B | 0.47 | 0.31 | -0.07 | 0.43 | 0.70/0.24 |
| | Full B → Full A | 0.58 | 0.56 | 0.18 | 0.64 | 0.62/0.54 |
| | Full A → Stroop | 0.42 | 0.34 | -0.18 | 0.33 | 0.65/0.20 |
| | Stroop → Full A | 0.52 | 0.43 | 0.07 | 0.62 | 0.13/0.91 |
| | Full A → Arithmetic | 0.53 | 0.45 | 0.05 | 0.56 | 0.60/0.45 |
| | Arithmetic → Full A | 0.59 | 0.57 | 0.20 | 0.66 | 0.68/0.50 |
| | Full A → Mirror | 0.47 | 0.39 | -0.05 | 0.43 | 0.66/0.29 |
| | Mirror → Full A | 0.54 | 0.50 | 0.08 | 0.57 | 0.76/0.31 |